Jiajia Li

Curriculum Vitæ

RESEARCH INTEREST

High-performance computing with a focus on the interaction among applications, numerical methods, algorithms, automatic performance tuning, performance profiling, program analysis, and computer architectures. I am particularly interested in building high-performance systems for sparse and irregular workloads, including sparse (multi-)linear algebra, solvers, tensor decompositions, and graph algorithms for large-scale data analytics and domain applications on diverse computer architectures.

EDUCATION

2013 – 2018 **Ph.D.**, *Georgia Institute of Technology*, Computational Science & Engineering, Advisor: Prof. Richard Vuduc.

2008 – 2013 **Doctor of Engineering**, *University of Chinese Academy of Sciences (UCAS)*, Computer Science, Advisors: Prof. Mingyu Chen and Guangming Tan.

2005 – 2008 **Bachelor of Sciences**, *Dalian University of Technology*, Computational Mathematics, In the Accelerated Student Program (2/180).

PROFESSIONAL EXPERIENCE

2022-Now **Assistant Professor**, *Computer Science, North Carolina State University (NCSU)*, Raleigh, NC.

2021-2022 **Assistant Professor**, *Computer Science, College of William & Mary (W&M)*, Williamsburg, VA.

2018-2021 **Research Scientist**, *HPC Group, Pacific Northwest National Laboratory (PNNL)*, Richland, WA.

2014-2018 **Graduate Research Assistant**, *HPC Garage, Georgia Institute of Technology*, Atlanta, GA. Advisor: Prof. Richard Vuduc.

2016 **Summer Research Intern**, *IBM Thomas J. Watson Research Center*, Mentors: Dr. Jee Choi and Dr. Dong Chen.

2015 **Summer Research Intern**, *Intel Parallel Computing Research Lab*, Mentor: Dr. Mikhail Smelyanskiy.

2013-2014 **Graduate Research Assistant**, *HPC LAB, Georgia Institute of Technology*, Atlanta, GA. Advisor: Prof. David Bader.

2008-2013 **Graduate Research Assistant**, *CARCH, Institute of Computing Technology Chinese Academy of Sciences*, Atlanta, GA. Advisors: Prof. Guangming Tan and Prof. Mingyu Chen.

HONORS AND AWARDS

2021 The 39th IEEE International Conference on Computer Design (ICCD'21) **Best Paper Award**

2019 Rising Stars for Women in Computational and Data Sciences. [Link]

2019 Principles and Practice of Parallel Programming. (PPoPP'19) **Best Paper Award Finalist**

2018 ACM/IEEE International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC'18) **Best Student Paper Award**.[Link][Link]

2018 SIAM ALA'18 Student Travel Grant.

2018 GaTech CoC Graduate Student Council Travel Grant.

2017 **IBM PhD Fellowship** for 2017-2018. [Link]

2017 Travel grant from ATIP Workshop, co-located with SC'17 [Link]

2017 Travel grant from IPAM for Big Data Meets Computation Workshop 2017 [Link]

2016 Selected students to attend IEEE WIE Women's Leadership Summit

2013 ZhuLiYueHua Award for the Excellent PhD Students of Chinese Academy of Sciences (Top 0.2%)
2011 Xia Peisu Scholarship of Institute of Computing Technology (Top 1%)
2011 Outstanding Research Assistant of the Computer Architecture Laboratory at UCAS
2010 Outstanding Student of the Computer Architecture Laboratory at UCAS

PUBLICATIONS

Students advised or co-advised by me are double-underlined; people mentored by me are underlined.

SC26 Zecheng Li, Sri Harshavardhan Reddy Deverapalli, Yanbo Zhao, Frank Mueller, Karl Pierce, **Jiajia Li**. TIDES: Tiered Block-Sparse Tensor Contraction with Streaming Transpose on GPUs. The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC). 2026. (Accepted)

OSDI26 Zecheng Li, Xu Liu, Namhyung Kim, Blake Jones, Alexey Alexandrov, **Jiajia Li**. TypeCraft: A Lightweight Data Type Profiler with High Resolution. USENIX Symposium on Operating Systems Design and Implementation (OSDI). 2026. (Accepted)

PLDI26 Zhen Du, Ying Liu, Xionghui Chen, Yanbo Zhao, Xiaobing Feng, Huimin Cui, **Jiajia Li**. SparseZETA: Intelligent Auto-Tuner for Designing High-Performance SpMV Programs. Programming Language Design and Implementation (PLDI). 2026. (Accepted)

ARRAY26 Yanbo Zhao, Zhaonan Meng, Sai Krishna Teja Varma Manthena, Xu Liu, Ajay Panyala, **Jiajia Li**. Leveraging AI Ecosystem for Portable and Sustainable GPU Kernels in HPC. ARRAY Workshop, co-located with Programming Language Design and Implementation (PLDI). 2026. (Accepted)

FSE26 Jinku Cui, Yueming Hao, Shuyin Jiao, **Jiajia Li**, Xu Liu. SmartDispatch: Dynamic Substitution of NumPy-style APIs on Heterogenous CPU-GPU Systems. Foundations of Software Engineering (FSE). 2026. (Accepted)

MLSys26 Ran Ran, Zhaoting Gong, Zhaowei Li, Xianting Lu, **Jiajia Li**, Wujie Wen. G-HEMP: Fast Multi-GPU Private Inference for Large-Scale GCNs with Homomorphic Encryption. MLSys. 2026. (Accepted)

IPDPS26 Zhaonan Meng, Miles Stoudenmire, Karl Pierce, Frank Mueller, **Jiajia Li**. STTID: High-Performance Sparse Tensor-Train Interpolative Decomposition. IEEE International Parallel & Distributed Processing Symposium (IPDPS). 2026.

ICS26 Srikar Chundury, Blake Burgstahler, **Jiajia Li**, In-Saeng Suh, Frank Mueller. Diagonal-Budgeted Trotterization for Efficient Quantum Hamiltonian Simulation. International Conference on Supercomputing (ICS). 2026. (Accepted)

ICPP26 Yongseok Soh, Shruti Shivakumar, **Jiajia Li**, Jee Choi, Ramakrishnan Kannan. From 2^N to N^2: Tree-Free Scalable Sparse Symmetric Tucker Decomposition. International Conference on Parallel Processing (ICPP). 2026. (Accepted)

TACO26 Guofeng Feng, Zecheng Li, Mingzhen Li, Weile Jia, Ninghui Sun, Guangming Tan, **Jiajia Li**. Bullseye Hash: An Efficient Hash-Table for Sparse Tensor Contraction. ACM Transactions on Architecture and Code Optimization (TACO). 2026. (Accepted)

ArXiv26 Zhaonan Meng, Yuehaw Khoo, **Jiajia Li**, E. Miles Stoudenmire. Recursive Sketched Interpolation: Efficient Hadamard Products of Tensor Trains. ArXiv preprint arXiv:2602.17974. 2026.

IPDPS25 Zecheng Li, Shruti Shivakumar, **Jiajia Li**, Ramakrishnan Kannan. SymProp: Scaling Sparse Symmetric Tucker Decomposition via Symmetry Propagation. IEEE International Parallel & Distributed Processing Symposium (IPDPS). 2025.

ASPLOS25 Qidong Zhao, Hao Wu, Yueming Hao, Zilingfeng Ye, **Jiajia Li**, Xu Liu, Keren Zhou. DeepContext: A Context-aware, Cross-platform, and Cross-framework Tool for Performance Profiling and Analysis of Deep Learning Workloads. Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Volume 3. 2025, pp. 48–63.

SC25 Yanbo Zhao, Yueming Hao, Zecheng Li, Shuyin Jiao, Xu Liu, **Jiajia Li**. RedSan: A Redundant Memory Instruction Sanitizer for GPU Programs. Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC). 2025, pp. 368–382.

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TACO25 Zhen Du, Ying Liu, Ninghui Sun, Huimin Cui, Xiaobing Feng, **Jiajia Li**. SRSparse: Generating Codes for High-Performance Sparse Matrix-Vector Semiring Computations. ACM Transactions on Architecture and Code Optimization 22, no. 2 (2025): 1-26.

TACO25 Zhenlin Wu, Tianao Ge, **Jiajia Li**, Xinyu Chen, Hongyuan Liu. Advancing Matrix Operations for High-Performance and Memory-Efficient Automata Processing on GPUs. ACM Transactions on Architecture and Code Optimization 22, no. 4 (2025): 1-26.

TACO25 Zhenlin Wu, Haosong Zhao, Hongyuan Liu, Wujie Wen, **Jiajia Li**. gHyPart: GPU-friendly End-to-End Hypergraph Partitioner. ACM Transactions on Architecture and Code Optimization. 2025.

ArXiv25 Yanbo Zhao, Jinku Cui, Zecheng Li, Shuyin Jiao, Xu Liu, **Jiajia Li**. cuThermo: Understanding GPU Memory Inefficiencies with Heat Map Profiling. ArXiv preprint arXiv:2507.18729. 2025.

ICS24 Keren Zhou, Karthik Ganapathi Subramanian, Po-Hsun Lin, Matthias Fey, Binqian Yin, **Jiajia Li**. FASTEN: Fast GPU-accelerated Segmented Matrix Multiplication for Heterogenous Graph Neural Networks. Proceedings of the 38th ACM International Conference on Supercomputing (ICS). 2024, pp. 511–524.

SC24 Zecheng Li, **Jiajia Li**. PINE: Efficient Yet Effective Piecewise Linear Trees. ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC). 2024. (Poster)

PPoPP24 Guofeng Feng, Weile Jia, Ninghui Sun, Guangming Tan, **Jiajia Li**. POSTER: Optimizing Sparse Tensor Contraction with Revisiting Hash Table Design. Principles and Practice of Parallel Programming (PPoPP). 2024. (Short Paper)

ICLR24-W Hongjue Zhao, Yuchen Wang, Hairong Qi, **Jiajia Li**, Lui Sha, Han Zhao, Huajie Shao. Accelerating Neural Differential Equations for Irregularly-Sampled Dynamical Systems Using Variational Formulation. ICLR Workshop on AI4DifferentialEquations In Science. 2024.

PPoPP23 Zhen Xie, Jie Liu, **Jiajia Li**, Dong Li. Merchandiser: Data Placement on Heterogeneous Memory for Task-Parallel HPC Applications with Load-Balance Awareness. Principles and Practice of Parallel Programming (PPoPP). 2023.

ToPC Zheng Miao, Jon Calhoun, Rong Ge, **Jiajia Li**. Performance Implication of Tensor Irregularity and Optimization for Distributed Tensor Decomposition. ACM Transactions on Parallel Computing. 2023.

HiPC23 Shruti Shivakumar, Ilya Amburg, Sinan G. Aksoy, **Jiajia Li**, Stephen J. Young, Srinivas Aluru. Fast Parallel Tensor Times Same Vector for Hypergraphs. IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC). 2023, pp. 324–334.

SC22 Zhen Du, **Jiajia Li**, Yinshan Wang, Xueqi Li, Guangming Tan, Ninghui Sun. AlphaSparse: Generating High Performance SpMV Codes Directly from Sparse Matrices. The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC). 2022.

CLUSTER22 Zheng Miao, **Jiajia Li**, Jon Calhoun, Rong Ge. BALA-CPD: BALanced and Asynchronous Distributed Tensor Decomposition. 24th IEEE Cluster. 2022.

HPCA22 Cheng Tan, Nicolas Bohm Agostini, Tong Geng, Chenhao Xie, **Jiajia Li**, Ang Li, Kevin Barker, Antonino Tumeo. DRIPS: Dynamic Rebalancing of Pipelined Streaming Applications CGRAs. The 28th IEEE International Symposium on High-Performance Computer Architecture (HPCA). 2022.

PPoPP22 Zhen Xie, Jie Liu, Yuchen Ma, **Jiajia Li**, Dong Li. LB-HM: Load Balance-Aware Data Placement on Heterogeneous Memory for Task-Parallel HPC Applications. Principles and Practice of Parallel Programming (PPoPP). 2022. (Poster)

LLVM-HPC21 Ruiqin Tian, Luanzheng Guo, **Jiajia Li**, Bin Ren, Gokcen Kestor. A High Performance Sparse Tensor Algebra Compiler in MLIR. The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), the Seventh Annual Workshop on the LLVM Compiler Infrastructure in HPC (LLVM-HPC). 2021.

ICCD21 Cheng Tan, Tong Geng, Chenhao Xie, Nicolas Bohm Agostini, **Jiajia Li**, Ang Li, Kevin Barker, Antonino Tumeo. DynPaC: Coarse-Grained, Dynamic, and Partially Reconfigurable Array for Streaming Applications. The 39th IEEE International Conference on Computer Design (ICCD). 2021. (AR=24.4%) (**Best Paper Award**)

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HPEC21 Tong Geng, Chunshu Wu, Cheng Tan, Chenhao Xie, Anqi Guo, Pouya Haghi, Sarah Yuan He, **Jiajia Li**, Martin Herbordt, Ang Li. A Survey: Handling Irregularities in Neural Network Acceleration with FPGAs. IEEE High Performance Extreme Computing Conference (HPEC). 2021.

ACDA21 Shruti Shivakumar, **Jiajia Li**, Ramakrishnan Kannan, Srinivas Aluru. Efficient Parallel Sparse Symmetric Tucker Decomposition for High-Order Tensors. SIAM Conference on Applied and Computational Discrete Algorithms (ACDA). 2021.

ICS21 Jiawen Liu, Dong Li, Roberto Gioiosa, **Jiajia Li**. Athena: High-Performance Sparse Tensor Contraction Sequenceon Heterogeneous Memory. International Conference on Supercomputing (ICS). 2021. (AR=25%)

PPoPP21 Jiawen Liu, Jie Ren, Roberto Gioiosa, Dong Li, **Jiajia Li**. Sparta: High-Performance, Element-Wise Sparse Tensor Contraction on Heterogeneous Memory. Principles and Practice of Parallel Programming (PPoPP). 2021. (AR=21%)

NVMW21 Jiawen Liu, Jie Ren, Roberto Gioiosa, Dong Li, **Jiajia Li**. Sparta: High-Performance, Element-Wise Sparse Tensor Contraction on Heterogeneous Memory. 12th Annual Non-Volatile Memories Workshop (NVMW). 2021.

IISWC20 **Jiajia Li**, Mahesh Lakshminarasimhan, Xiaolong Wu, Ang Li, Catherine Olschanowsky, Kevin Barker. A Sparse Tensor Benchmark Suite for CPUs and GPUs. IEEE International Symposium on Workload Characterization (IISWC). 2020.

MLSH20 Xiaolong Wu, Yang Yi, Dave (Jing) Tian, **Jiajia Li**. Generic, Sparse Tensor Core for Neural Networks. 1st International Workshop on Machine Learning for Software Hardware Co-Design (MLSH), in conjunction with the 29th International Conference on Parallel Architectures and Compilation Techniques (PACT). 2020.

ToPC Eric Hein, Srinivas Eswar, Abdurrahman Yasar, **Jiajia Li**, Jeffrey S. Young, Tom Conte, Umit V. Catalyurek, Rich Vuduc, Jason Riedy, Bora Ucar. Programming Strategies for Irregular Algorithms on the Emu Chick. ACM Transactions on Parallel Computing. 2020.

SC20-Poster Zheng Miao, Jon C. Calhoun, Rong Ge, **Jiajia Li**. Sparsity-Aware Distributed Tensor Decomposition. ACM/IEEE International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC). 2020. (Poster)

WHPC20 Ruiqin Tian, **Jiajia Li**, Bin Ren, Gokcen Kestor. High-Performance Sparse Tensor Algebra Compiler. Women in High Performance Computing Workshop (WHPC), in conjunction with ACM/IEEE International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC). 2020. (Poster)

HPEC20 Jesun Sahariar Firoz, Ang Li, **Jiajia Li**, Kevin Barker. On the Feasibility of Using Reduced-PrecisionTensor Core Operations for Graph Analytics. IEEE High Performance Extreme Computing Conference (HPEC). 2020.

PPoPP20 **Jiajia Li**, Mahesh Lakshminarasimhan, Xiaolong Wu, Ang Li, Catherine Olschanowsky, Kevin Barker. A Parallel Sparse Tensor Benchmark Suite on CPUs and GPUs. Principles and Practice of Parallel Programming (PPoPP). 2020. (Poster)

TPDS Ang Li, Shuaiwen Leon Song, Jieyang Chen, **Jiajia Li**, Xu Liu, Nathan Tallent, Kevin Barker. Evaluating Modern GPU Interconnect: PCIe, NVLink, NV-SLI, NVSwitch and GPUDirect. IEEE Transactions on Parallel and Distributed Systems. 2020.

SC19 Israt Nisa, **Jiajia Li**, Aravind Sukumaran-Rajam, Prashant Rawat, Sriram Krishnamoorthy, P. (Saday) Sadayappan. An Efficient Mixed-Mode Representation of Sparse Tensors. ACM/IEEE International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC). 2019.

ICS19 **Jiajia Li**, Bora Ucar, Umit Catalyurek, Jimeng Sun, Kevin Barker, Richard Vuduc. Efficient and Effective Sparse Tensor Reordering. International Conference on Supercomputing (ICS). 2019.

THPC **Jiajia Li**, Yuchen Ma, Xiaolong Wu, Ang Li, Kevin Barker. PASTA: A Parallel Sparse Tensor Algorithm Benchmark Suite. CCF Transactions on High Performance Computing. 2019.

ParCo Jeffrey S. Young, Eric Hein, Srinivas Eswar, Patrick Lavin, **Jiajia Li**, Jason Riedy, Richard Vuduc, Thomas M. Conte. A Microbenchmark Characterization of the Emu Chick. Journal of Parallel Computing. 2019.

PPoPP19 Ke Meng, **Jiajia Li**, Guangming Tan. A Pattern Based Algorithmic Autotuner for Graph Processing on GPUs. Principles and Practice of Parallel Programming (PPoPP). 2019. (**Best Paper Award Finalist**)

IPDPS19 Israt Nisa, **Jiajia Li**, Aravind Sukumaran Rajam, Richard Vuduc, P. (Saday) Sadayappan. Load-balanced sparse MTTKRP on GPUs. IEEE International Parallel & Distributed Processing Symposium (IPDPS). 2019.

TOPC Junhong Liu, Guangming Tan, Yulong Luo, **Jiajia Li**, Zeyao Mo, Ninghui Sun. An Autotuning Protocol to Rapidly Build Autotuners. ACM Transactions on Parallel Computing. 2019.

THESIS **Jiajia Li**. Scalable Tensor Decompositions in High Performance Computing Environments. PhD Dissertation. Georgia Institute of Technology, Atlanta, GA, USA. July 2018.

SC18 **Jiajia Li**, Jimeng Sun, Richard Vuduc. HiCOO: Hierarchical Storage of Sparse Tensors. ACM/IEEE International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC). 2018. (**Best Student Paper Award**)

JPDC Yuchen Ma, **Jiajia Li**, Xiaolong Wu, Chenggang Yan, Jimeng Sun, Richard Vuduc. Optimizing Sparse Tensor Times Matrix on GPUs. Journal of Parallel and Distributed Computing (Special Issue on Systems for Learning, Inferencing, and Discovering). 2018.

IPDPSW18 Eric Hein, Tom Conte, Jeffrey Young, Srinivas Eswar, **Jiajia Li**, Patrick Lavin, Richard Vuduc, Jason Riedy. An Initial Characterization of the Emu Chick. 2018 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW). 2018.

PPoPP18 Yue Zhao, **Jiajia Li**, Chunhua Liao, Xipeng Shen. Bridging the Gap between Deep Learning and Sparse Matrix Format Selection. 23rd ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP). 2018.

TOMS Guangming Tan, Junhong Liu, **Jiajia Li**. Design and Implementation of Adaptive SpMV Library for Multicore and Manycore Architecture. ACM Transactions on Mathematical Software. 2018.

IPDPS17 **Jiajia Li**, Jee Choi, Ioakeim Perros, Jimeng Sun, Richard Vuduc. Model-Driven Sparse CP Decomposition for Higher-Order Tensors. 31st IEEE International Parallel & Distributed Processing Symposium (IPDPS). 2017.

PACT17 Yue Zhao, **Jiajia Li**, Chunhua Liao, Xipeng Shen. Bridging the Gap between Deep Learning and Sparse Matrix Format Selection. The 26th International Conference on Parallel Architectures and Compilation Techniques (PACT). 2017. (Poster)

PPoPP17 Xiuxia Zhang, Guangming Tan, Shuangbai Xue, **Jiajia Li**, Keren Zhou, Mingyu Chen. Understanding the GPU Microarchitecture to Achieve Bare-Metal Performance Tuning (PPoPP). 2017. (**Best Artifact Award**)

SC16-IA3 **Jiajia Li**, Yuchen Ma, Chenggang Yan, Richard Vuduc. Optimizing Sparse Tensor Times Matrix on multi-core and many-core architectures. The sixth Workshop on Irregular Applications: Architectures and Algorithms (IA^3), co-located with SC. 2016.

SC15 **Jiajia Li**, Casey Battaglino, Ioakeim Perros, Jimeng Sun, Richard Vuduc. An Input-Adaptive and In-Place Approach to Dense Tensor-Times-Matrix Multiply. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC) 2015

SC15-EduHPC B. Neelima, **Jiajia Li**. Introducing high performance computing concepts into engineering undergraduate curriculum: a success story. Proceedings of the Workshop on Education for High-Performance Computing (EduHPC), co-located with SC. 2015

THESIS **Jiajia Li**. Research on Sparse Matrix Vector Multiplication Auto-tuning Method. PhD Thesis. The University of Chinese Academy of Sciences, Beijing, China. July, 2013

PLDI13 **Jiajia Li**, Guangming Tan, Mingyu Chen, Ninghui Sun. SMAT: An Input Adaptive Auto-Tuner for Sparse Matrix-Vector Multiplication. Programming Language Design and Implementation (PLDI) 2013

ICS12 **Jiajia Li**, Xingjian Li, Guangming Tan, Mingyu Chen, Ninghui Sun. An Optimized Large-Scale Hybrid DGEMM Design for CPUs and ATI GPUs. International Conference on Supercomputing (ICS) 2012

JCRD **Jiajia Li**, Xiuxia Zhang, Guangming Tan, Mingyu Chen. Study of Choosing the Best Storage Format of Sparse Matrix Vector Multiplication, Journal of Computer Research and Development. (IN CHINESE) 2012

ICPADS10 **Jiajia Li**, Guangming Tan, Mingyu Chen. Automatically Tuned Dynamic Programming with an Algorithm-by-Blocks. The 16th International Conference on Parallel and Distributed Systems (ICPADS) 2010

GRANTS

As *Principal Investigator*

Ragged **Jiajia Li** (PI), Keren Zhou (Lead, GMU). Collaborative Research: SHF: Hierarchical Optimization of Ragged Tensor Operators for Deep Learning Workloads. NSF #2554013. Estimated Award: $268,768 (NCSU portion). Jul. 2026 - Jun. 2029.

CS2 **Jiajia Li** (Lead), Bo Yuan, Cunxi Yu, Ajay Panyala. Collaborative Research: CS2: Formally Verified and Performance-Optimized Tensor Contraction Sequences in Quantum Many-Body Computations. NSF #2546785. Estimated Award: $1.5M (NCSU portion $267K). May 2026 - Apr. 2029.

CROSS-Large **Jiajia Li** (Lead), Frank Mueller, Dong Li, Lizhong Chen. Collaborative Research: PPoSS: LARGE: Cross-layer Coordination and Optimization for Scalable and Sparse Tensor Networks (CROSS). NSF #2316201. Estimated Award: $5M (NCSU portion $3,033,782). Sep. 2023 - Aug. 2028.

DrGPU **Jiajia Li**, Xu Liu. Collaborative Research: CNS Core: SMALL: DrGPU: Optimizing GPU Programs via Novel Profiling Techniques. NSF #2125732. Estimated Award: $250K. Oct. 2021 - Sep. 2025.

CROSS-Plan **Jiajia Li** (Lead), Frank Mueller, Dong Li, Lizhong Chen. Collaborative Research: PPoSS: Planning: Cross-layer Coordination and Optimization for Scalable and Sparse Tensor Networks (CROSS). NSF #532355. Estimated Award: $250K (NCSU portion $125K). Oct. 2022 - Sep. 2024.

CloudSvc **Jiajia Li**, Xu Liu. Collaborative Research: CNS Core: Small: Towards Efficient Cloud Services. NSF #2006373. Estimated Award: $266K. Oct. 2020 - Sep. 2024.

SHARWK Stephen J. Young (Lead), Sinan G. Aksoy, **Jiajia Li** (Co-PI). SHARWK: Scalable Hypergraph Analysis Via Random Walk Kernels. DOE EXPRESS: Randomized Algorithms for Extreme-Scale Science. Estimated Award: $78,382 (NCSU portion). Apr. 2022 - Sep. 2023.

HiParTI **Jiajia Li** (Lead), Ang Li, Ajay Panyala (Key Staffs). Application-Algorithm-Architecture Co-Design for Large-Scale, Sparse Tensor/Matrix Methods. PNNL Data-Model Convergence LDRD. Estimated Award: $630K. Aug. 2019 - Oct. 2022.

As *Key Staff*

CENATE Kevin Barker (PI), **Jiajia Li** (Key Staff of Tensor Task), et al. The Center for Advanced Technology Evaluation. DOE ASCR No. 66150. Oct. 2019 - Oct. 2022.

NWChemEx Thom Dunning Jr, **Jiajia Li** (Key Staff of TAMM task, led by Sriram Krishnamoorthy), et al. Tackling Chemical, Materials, and Biomolecular Challenges in Exascale. DOE ECP Project. Oct. 2016 - Oct. 2023.

ExaSGD Henry Huang (PI), **Jiajia Li** (Key Staff in GPU thrust, led by Slaven Peles), et al. Optimizing Stochastic Grid Dynamics at Exascale. DOE ECP Project. Oct. 2019 - Oct. 2020.

CFA Ang Li (PI), Xiu Yang, Leon Song, **Jiajia Li** (Key Staff), Jesun Firoz. Computation-Flow-Architecture: Designing Non-Von-Neumann Architecture for Future Data-Centric Computing. PNNL Data-Model Convergence LDRD. Estimated Award: $620K. Aug. 2019 - Oct. 2022.

S-BLAS Ang Li (PI), **Jiajia Li** (Key Staff), Jesun Firoz. Designing Highly Scalable Sparse-BLAS Kernels for Complex Modern HPC Architectures. PNNL High Performance Data Analytics Program. Awarded: $170K. Mar. 2019 - Mar. 2020.

SOFTWARE

HiParTI **A Hierarchical Parallel Tensor Infrastructure**, [Link].

PASTA **A Parallel Sparse Tensor Algorithm Benchmark Suite**, [Link].

ParTI! **A Parallel Tensor Infrastructure for Data Analysis**, [Link].

AdaTM **Adaptive Tensor Memoization Algorithm for CP Decomposition**, [Link].

InTensLi **Input-Adaptive and In-Place Dense Tensor-Times-Matrix Multiply**, [Link].

SMAT **SpMV Auto-Tuner**, [Link].

HDGEMM **A Hybrid DGEMM library on a Heterogeneous CPU-AMD GPU Architecture**, [Link].

INVITED TALKS AND PRESENTATIONS

2026 **High-Performance Block-Sparse Tensor Contractions on GPUs**, *(Invited). SIAM Conference on Parallel Processing for Scientific Computing (SIAM-PP'26)*, Berlin, Germany, March, 2026.

2025 **Intelligent Auto-Tuning for High-Performance Sparse Tensor Algebra**, *(Invited). 46th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI), the Sparse workshop (PLDI-Sparse'25)*, Seoul, South Korea, June, 2025.

2024 **Discussion on Hash-Table Approaches for Efficient Sparse Tensor Contraction**, . *ACM/IEEE International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC), 14th Workshop on Irregular Applications: Architectures and Algorithms (SC-IA'24)*, Atlanta, GA, Nov, 2024.

2021 **How to Do Successful Career Change**, *(Invited). ACM/IEEE International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC'21) Women in HPC workshop (WHPC)*, Online, Nov, 2021.

High-Performance Sparse Tensor Operations in HiParTI Library, *(Invited). CCF HPC China 2021*, Online, Oct, 2021.

Sparsity-Aware Distributed Tensor Decomposition, *(Invited). 2021 SIAM Conference on Applied Linear Algebra (SIAM-LA21)*, New Orleans, Louisiana, May, 2021.

High-Performance Sparse Tensor Operations in HiParTI Library, *(Invited). SPCL_Bcast organized by Scalable Parallel Computing Laboratory (SPCL), ETH*, Online, Feb 25, 2021.

Tensor Methods and Emerging Applications to the Physical and Data Sciences, *(Invited). 2021 IPAM Long Program (TM2021)*, Los Angeles, CA, Mar-Jun, 2021.

Sparta: High-Performance, Element-Wise SparseTensor Contraction on Heterogeneous Memory, *(Invited). 2021 SIAM Conference on Computational Science and Engineering (SIAM-CSE21)*, Fort Worth, TX, March, 2021.

2020 **PASTA: A Parallel Sparse Tensor Algorithm Benchmark Suite**, *(Invited). SIAM Conference on Parallel Processing for Scientific Computing (SIAM-PP20)*, Seattle, WA, February, 2020.

2019 **Sparse Tensor Library based on HiCOO Format**, *(Invited). AI and Tensor Factorization for Physical, Chemical, and Biological Systems Workshop*, Santa Fe, NM, September, 2019.

Efficient and Effective Sparse Tensor Reordering, *(Paper). International Conference on Supercomputing (ICS19)*, Phoenix, AZ, June, 2019.

Sparse Tensor Algebra and its Relations to Matrix and Graph Problems, *(Invited). SIAM Conference on Computational Science and Engineering (SIAM-CSE19)*, Spokane, WA, February, 2019.

A Sparse Tensor Format and a Benchmark Suite, *(Invited). Workshop on Compiler Techniques for Sparse Tensor Algebra*, Cambridge, MA, January, 2019.

2018 **HiCOO: Hierarchical Storage of Sparse Tensors**, *(Paper). ACM/IEEE International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC18)*, Dallas, TX, November, 2018.

Scalable Tensor Decompositions in High Performance Computing Environments, *Ph.D. Thesis Defense*, Atlanta, GA, July, 2018.

ParTI!: A Parallel Tensor Infrastructure on HPC Platforms, *(Invited). The 8th edition of the multidisciplinary conference on ThRee-way methods In Chemistry And Psychology (TRICAP18)*, Angel Fire, NM, June, 2018.

Parallel Sparse Tensor Decompositions using HiCOO Format, *(Invited). SIAM Conference on Applied Linear Algebra (SIAM-ALA18)*, Hong Kong, May, 2018.

HiCOO: Hierarchical Storage of Sparse Tensors, *(Invited). SIAM Conference on Parallel Processing for Scientific Computing (SIAM-PP18)*, Tokyo, Japan, March, 2018.

2017 **HiCOO: A Hierarchical Sparse Tensor Format for Tensor Decompositions**, *(Poster). The International Conference for High Performance Computing, Networking, Storage and Analysis (SC17)*, Denver, CO, November, 2017..

ParTI!: A Parallel Tensor Infrastructure, *The International Conference for High Performance Computing, Networking, Storage and Analysis (SC17) ATIP workshop on International Exascale and Next-Generation Computing Programs*, Denver, CO, November, 2017.

Model-Driven Sparse CP Decomposition for Higher-Order Tensors, *(Invited). SIAM Annual Meeting (SIAM-AN17)*, Pittsburgh, PA, July, 2017.

Model-Driven Sparse CP Decomposition for Higher-Order Tensors, *(Paper). The 31st IEEE International Parallel & Distributed Processing Symposium (IPDPS17)*, Orlando, FL, May, 2017.

Non-negative Sparse Tensor Decomposition on Distributed Memory Systems, *(Poster). SIAM Conference on Computational Science and Engineering (SIAM-CSE17)*, Atlanta, GA, February, 2017.

ParTI!: A Parallel Tensor Infrastructure for Data Analysis, *(Poster). Institute for Pure & Applied Mathematics Big Data Meets Computation Workshop (IPAM-DMC17)*, Los Angeles, CA, January, 2017.

2016 **Optimizing Sparse Tensor Times Matrix on Multi-core and Many-core Architectures**, *(Paper). The International Conference for High Performance Computing, Networking, Storage and Analysis (SC16), the sixth Workshop on Irregular Applications: Architectures and Algorithms (IA^3)*, Salt Lake City, UT, November, 2016.

Model-driven Sparse CP Decomposition for High-Order Tensors, *(Poster). The International Conference for High Performance Computing, Networking, Storage and Analysis (SC16), Women in HPC: Diversifying the HPC Community (WHPC) Workshop*, Salt Lake City, UT, November, 2016.

ParTI!: A Parallel Tensor Infrastructure for Data Analysis, *(Paper). The Conference on Neural Information Processing Systems, the Workshop on Learning with Tensors: Why Now and How? (NeurIPS16 Tensor-Learn)*, Barcelona, Spain, December, 2016.

An Input-Adaptive and In-Place Approach to Dense Tensor-Times-Matrix Multiply, *(Invited). SIAM Conference on Parallel Processing for Scientific Computing (SIAM-PP16)*, Paris, France, April, 2016.

2015 **An Input-Adaptive and In-Place Approach to Dense Tensor-Times-Matrix Multiply**, *(Paper). The International Conference for High Performance Computing, Networking, Storage and Analysis (SC15)*, Austin, TX, November, 2015.

Introducing high performance computing concepts into engineering undergraduate curriculum: a success story, *(Paper). The International Conference for High Performance Computing, Networking, Storage and Analysis (SC15), the Workshop on Education for High-Performance Computing (EduHPC)*, Austin, TX, November, 2015.

2013 **Research on Sparse Matrix Vector Multiplication Auto-tuning Method**, *Ph.D. Thesis Defense*, Beijing, China, July, 2013.

SMAT: An Input Adaptive Auto-Tuner for Sparse Matrix-Vector Multiplication, *(Paper). Programming Language Design and Implementation (PLDI13)*, Seattle, WA, June, 2013.

2012 **An Optimized Large-Scale Hybrid DGEMM Design for CPUs and ATI GPUs**, *(Paper). International Conference on Supercomputing (ICS12)*, Venice, Italy, June, 2012.

2010 **Automatically Tuned Dynamic Programming with an Algorithm-by-Blocks**, *(Paper). The 16th International Conference on Parallel and Distributed Systems (ICPADS10)*, Shanghai, China, December, 2010.

ORGANIZING ACTIVITIES

2026 BigData PC Vice Chair for "Big Data Infrastructure"

AI & ML Track PC Chair of the International Supercomputing Conference (ISC'26)

Artifact Evaluation Chair of the Principles and Practice of Parallel Programming Conference (PPoPP'26)

2025 Finance Chair of the International Conference on Parallel Architectures and Compilation Techniques (PACT'25)

Artifact Evaluation Chair of the Principles and Practice of Parallel Programming (PPoPP'25)

2024 Registration Chair of the International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'24)

General Co-Chair of the 1st Workshop on Cross-stack Optimization of Tensor Methods (XTensor), held in conjunction with the International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'24)

Algorithms/Applications Track Co-Chair of International Conference on High Performance Computing in Asia-Pacific Region (HPC-Asia'24)

2023 Artifact Program Co-Chair of the International Conference on Compiler Construction (CC'23)

Industry Chair of the Principles and Practice of Parallel Programming (PPoPP'23)

2022 Program Chair of the Emerging Parallel and Distributed Runtime Systems and Middleware Workshop (IPDRM), held in conjunction with IEEE/ACM International Conference on High Performance Computing, Networking, Storage and Analysis (SC'22) https://ipdrm.github.io

2021 Program Co-Chair of the 26th International Workshop on High-Level Parallel Programming Models and Supportive Environments (HIPS'21)

Finance Chair of International Conference on Supercomputing (ICS'21) https://ics21.github.io/

Session Chair of ACM Capital Region Celebration of Women in Computing (CAPWIC'21) https://capwic.org/conferences/2021/

Artifact Evaluation Chair of the 2021 edition of the International Conference on Languages Compilers, Tools and Theory of Embedded Systems (LCTES'21) https://pldi21.sigplan.org/home/LCTES-2021

2020 Publicity Chair of International Conference on Parallel Architectures and Compilation Techniques (PACT'20) https://pact20.cc.gatech.edu

Finance and Session Chair of the Principles and Practice of Parallel Programming 2020 (PPoPP'20) https://ppopp20.sigplan.org

Publicity Chair of the 17th Annual IFIP International Conference on Network and Parallel Computing (NPC'20) http://www.ncic.ac.cn/npc2020

Publicity Chair of the Emerging Parallel and Distributed Runtime Systems and Middleware Workshop (IPDRM), held in conjunction with IEEE/ACM International Conference on High Performance Computing, Networking, Storage and Analysis (SC'20) https://ipdrm.github.io

2019 Proceeding Chair of the Emerging Parallel and Distributed Runtime Systems and Middleware Workshop (IPDRM), held in conjunction with IEEE/ACM International Conference on High Performance Computing, Networking, Storage and Analysis (SC'19) https://ipdrm.github.io

Co-Chair of the 25th International European Conference on Parallel and Distributed Computing (Euro-Par'19) https://2019.euro-par.org

Web and Social Media Chair of International Conference on Parallel Architectures and Compilation Techniques (PACT'19) https://pactconf.org

Co-Chair of The First International Workshop on the Intersection of High Performance Computing and Machine Learning (HPCaML'19), held in conjunction with International Symposium on Code Generation and Optimization (CGO'19) http://hpc.pnl.gov/hpcaml19/

Co-Organizer of SIAM Conference on Computational Science and Engineering (SIAM CSE'19) Minisymposium "High Performance Sparse Matrix, Tensor, and Graph Kernels" http://hpc.pnl.gov/siamcse19/

PEER REVIEW ACTIVITIES

2027 PC member of the International Conference on Database Systems for Advanced Applications (DASFAA'27), and the ACM International Conference on Web Search and Data Mining (WSDM'27).

2026 PC member of the Design Automation Conference (DAC'26), the International Symposium on High-Performance Parallel and Distributed Computing (HPDC'26), the International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'26), the IEEE International Conference on Big Data (BigData'26), the International Conference on Supercomputing (ICS'26), the IEEE International Parallel & Distributed Processing Symposium (IPDPS'26), the ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP'26), the International Conference on Parallel Processing (ICPP'26), the IEEE International Conference on Cluster Computing (CLUSTER'26), the IEEE International Conference on Data Mining (ICDM'26), the International Symposium on Code Generation and Optimization (CGO'26), the Workshop on Irregular Applications: Architectures and Algorithms (IA^3'26), and the Workshop on Artificial Intelligence and Machine Learning for Scientific Applications (AI4S'26).

2025 PC member of the ACM Symposium on Parallelism in Algorithms and Architectures (SPAA'25), the IEEE International Symposium on Cluster, Cloud, and Internet Computing (CCGrid'25), the International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'25), the IEEE International Conference on Big Data (BigData'25), the International Conference on Supercomputing (ICS'25), the IEEE International Parallel & Distributed Processing Symposium (IPDPS'25), the ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP'25), the Workshop on Artificial Intelligence and Machine Learning for Scientific Applications (AI4S'25), and the Workshop on Irregular Applications: Architectures and Algorithms (IA^3'25).

2024 PC member of the International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'24), the International Conference on Parallel Processing (ICPP'24), the SIAM Conference on Parallel Processing for Scientific Computing (SIAM PP'24), the Workshop on Artificial Intelligence and Machine Learning for Scientific Applications (AI4S'24), and the Workshop on Irregular Applications: Architectures and Algorithms (IA^3'24).

2023 PC member of the International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'23), the International Conference on Supercomputing (ICS'23), the IEEE International Parallel & Distributed Processing Symposium (IPDPS'23), the International Conference on Parallel Processing (ICPP'23), the IEEE International Conference on Cluster Computing (CLUSTER'23), the International Supercomputing Conference (ISC'23), and the IEEE International Conference on Distributed Computing Systems (ICDCS'23).

2022 PC member of the International Conference on Supercomputing (ICS'22), the International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'22), the IEEE International Parallel & Distributed Processing Symposium (IPDPS'22), the International Supercomputing Conference (ISC'22), the IEEE International Conference on Cluster Computing (CLUSTER'22), the ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP'22), and the SIAM Conference on Parallel Processing for Scientific Computing (SIAM PP'22).

2021 PC member of the International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'21), the IEEE Transactions on Parallel and Distributed Systems special session on "Parallel and Distributed Computing Techniques for AI, ML and DL", the International Conference on Languages Compilers, Tools and Theory of Embedded Systems (LCTES'21), the International Conference on Parallel Processing (ICPP'21), the IEEE International Conference on Cluster Computing (CLUSTER'21), the International Conference on Supercomputing (ICS'21), and the IEEE International Conference on Distributed Computing Systems (ICDCS'21); Review Editor on Journal of Applied Sciences.

2020 PC member of the IEEE International Parallel & Distributed Processing Symposium (IPDPS'20), the International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'20), the IEEE Transactions on Parallel and Distributed Systems special session on "Parallel and Distributed Computing Techniques for AI, ML and DL", the ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP'20), the International Conference on Parallel Processing (ICPP'20), the ACM Student Research Competition at the International Conference on Parallel Architectures and Compilation Techniques (PACT'20 SRC), the Workshop on Machine Learning in High Performance Computing Environments (MLHPC'20), the Annual IFIP International Conference on Network and Parallel Computing (NPC'20), the Research Poster Committee of the International Supercomputing Conference (ISC'20), and the MLBench workshop at the IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS'20); Review Editor on Frontiers in Applied Mathematics and Statistics and Algorithms.

2019 Technical Program Committee Member of the Algorithms track of the 26th IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC'19)

Travel Grant PC Member of the ACM Symposium on High-Performance Parallel and Distributed Computing 2019 (HPDC'19)

PC Member of the Workshop on Tensor Methods for Emerging Data Science Challenges (TMEDSC), held in conjunction with the 25th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD'19)

PC member of the 25th International European Conference on Parallel and Distributed Computing (Euro-Par'19)

PC member of the First International Workshop on the Intersection of High Performance Computing and Machine Learning (HPCaML'19), held in conjunction with International Symposium on Code Generation and Optimization (CGO'19) http://hpc.pnl.gov/hpcaml19/.

2017-2019 PC Member of the International Conference on Advanced Engineering Computing and Applications in Sciences (ADVCOMP'19)

2013-2019 PC member of Parallel Algorithm Track of "National Annual Conference on High Performance Computing (HPC China)".

2018 External PC member of The 32nd ACM International Conference on Supercomputing (ICS'18).

PC Member of the 33rd IEEE International Parallel & Distributed Processing Symposium (IPDPS'18).

2017 PC member of Student Research Competition (SRC) of The 23rd ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'18).

2014-Now Reviewer of the Transactions on Parallel and Distributed Systems (TPDS), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), ACM Transactions on Knowledge Discovery from Data, Transactions on Knowledge and Data Engineering (TKDE), Transactions on Computers, ACM Transactions on Mathematical Software (TOMS), Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), IEEE Access, Journal of Parallel and Distributed Computing (JPDC), ACM Transactions on Parallel Computing (TOPC), Parallel Computing Journal (ParCo), CCF Transactions on High Performance Computing (THPC), Frontiers of Computer Science, Algorithmica Journal, Journal of Low Power Electronics and Applications, Songklanakarin Journal of Science and Technology (SJST), International Journal of High Performance Computing Applications, the 47th International Conference on Parallel Processing (ICPP'18), the 48th International Conference on Parallel Processing (ICPP'19), the 21st IEEE International Conference on Parallel and Distributed Systems (ICPADS'15), the 2020 International Conference on Memory Systems (MEMSYS'20), the 29th ACM International Conference on Information and Knowledge Management (CIKM'20), ACM Transactions on Embedded Computing Systems (TECS), ACM Transactions on Architecture and Code Optimization (TACO)

OTHER ACTIVITIES

2024-2026 Computing Equipment and Infrastructure Task Force of Computer Science Department at NCSU.

2021-2022 Colloquium Committee Member at W&M.

2021-2022 Awards and Prizes Committee Member at W&M.

2019-2020 STEM Ambassador of the Office of STEM Education at PNNL.

2014-2018 Organizer of Hot CSE seminar, a PhD academic seminar in GT CSE.

2013-2017 Volunteer librarian of Repetitive Stress Injury (RSI) Lending Library of GT College of Computing.

Apr 2016 Volunteer judge for Undergraduate Research Opportunities Program 11th Annual Undergraduate Research Spring Symposium.

2013-2015 Volunteer reviewer of "President's Undergraduate Research Awards (PUMA)" and "National Center for Women & IT (NCWIT) Award"

2008-2009 Vice Minister of Academic Study of Student Union at UCAS

MENTORING EXPERIENCE

Ph.D. Students
- Devadatta Mandaogane, NCSU, USA. Fa. 2025-Now
- Feiyang Zheng, NCSU, USA. Fa. 2025-Now
- Sai Krishna Teja Varma Manthena, NCSU, USA. Fa. 2024-Now
- Zhaonan Meng, NCSU, USA. Fa. 2024-Now
- Rahmy Salman, NCSU, USA. Fa. 2024-Now
- Zecheng Li, NCSU, USA. Fa. 2023-Now
- Yanbo Zhao, NCSU, USA. Fa. 2022-Now (co-advise)
- Sogolsadat Mansouri, NCSU, USA. Fa. 2025-Now
- Yi Wang, NCSU, USA. Fa. 2021-Now (co-advise)
- Jinku Cui, NCSU, USA. Fa. 2020-Now (co-advise)
- Qidong Zhao, NCSU, USA. Fa. 2019-2025 (co-advised), now at Google
- Zhen Du, University of Chinese Academy of Sciences, China. 2021-2024 (mentored)
- Zhenlin Wu, Hong Kong University of Science and Technology (Guangzhou), China. 2022-2024 (mentored)
- Guofeng Feng, University of Chinese Academy of Sciences, China. 2022-2024 (mentored)
- Shruti Shivakumar, Georgia Institute of Technology, USA. 2020-2024 (mentored)
- Zheng Miao, Clemson University, USA. 2019-2022 (mentored)
- Jiawen Liu, University of California Merced, USA. 2020-2021 (mentored)
- Israt Nisa, Ohio State University, USA. 2018-2020 (mentored)
- Ke Meng, University of Chinese Academy of Sciences, China. 2018-2019 (mentored)
- Xiuxia Zhang, University of Chinese Academy of Sciences, China. 2016 (mentored)
- Subramanya Dulloor, Georgia Institute of Technology, USA. 2015 (mentored)

Masters
- Zizhong Wang, NCSU, USA. Fa. 2025-Now
- Sri Harshavardhan Reddy Deverapalli, NCSU, USA. graduated 2026, now at TensorQ
- Sai Krishna Teja Varma Manthena, NCSU, USA. 2022-2024 (Graduated)
- Ahmed Taimoor, NCSU, USA. 2023-2025 (Graduated)
- Mushtaq Ahmed Shaikh, NCSU, USA. 2025 (Graduated)
- Devadatta Mandaogane, NCSU, USA. 2025 (Graduated)
- Swarnamalya Mohan, NCSU, USA. 2023-2024 (Graduated)
- Sounder Rajendran, NCSU, USA. 2023-2024 (Graduated), now at AMD
- Karthik Ganapathi Subramanian, NCSU, USA. 2023-2024 (Graduated)
- Po-Hsun Lin, NCSU, USA. 2023-2024 (Graduated)
- Elijah Mas, William & Mary, USA. 2021-2022 (Graduated)
- Junghyun Kim, Georgia Institute of Technology, USA. 2017-2018 (Graduated)

Undergraduates
- Yuchen Ma, Hangzhou Dianzi University, China. 2016-2018 (Graduated)
- Nicholas Liu, Georgia Institute of Technology, USA. 2017 (Graduated)

TEACHING EXPERIENCE

- Fall 2025 Instructor of "[Grad] Architecture of Parallel Computers (CSC/ECE 506-01)" at NCSU (Enrollment: 40)
- Fall 2025 Instructor of "[Grad] Parallel Algorithms (CSC 491-005, CSC 591-126)" at NCSU (Enrollment: 8)
- Spring 2025 Instructor of "[Grad] Parallel Systems (CSC 548-01, ECE 591-029)" at NCSU (Enrollment: 62)
- Fall 2024 Instructor of "[Grad] Seminar in Computer Science (CSC 801-002)" at NCSU (Enrollment: 16+)
- Fall 2024 Instructor of "[Grad] Parallel Algorithms (CSC 591/791-126, ECE 591-025)" at NCSU (Enrollment: 19)
- Spring 2024 Instructor of "[Grad] Accelerating Deep Learning (CSC 495-004/591-104)" at NCSU (Enrollment: 26)

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Fall 2023 Instructor of "[Grad] Efficient Tensor Computation for AI and Scientific Applications (CSC 591/791-26)" at NCSU (Enrollment: 13)

Spring 2023 Instructor of "[Grad] Parallel Systems (CSC 548-01)" at NCSU (Enrollment: 15)

Spring 2023 Instructor of "[Grad] Accelerating Deep Learning (CSC 591/791-25)" at NCSU (Enrollment: 10)

Spring 2022 Instructor of "[Undergrad] Algorithms (CSCI 303-01)" at W&M (Enrollment: 62)

Fall 2021 Instructor of "[Grad] Accelerating Deep Learning (CSCI 780-02)" at W&M (Enrollment: 9)

Spring 2017 Teaching Assistant of "Intro to High-Performance Computing (OMSCS) (CSE 6220)" at Georgia Tech

Fall 2014 Teaching Assistant of "High-Performance Computing: Tools and Applications (CSE 6230)" at Georgia Tech

May 2012 Teaching Assistant of "Parallel Computer Architecture" class of Dragonstar Project at University of Chinese Academy of Sciences

Jun 2012 Instructor of "Parallel Computing on GPUs using CUDA" Training at Sun Yat-sen University

PROFESSIONAL MEMBERSHIPS

Member of the Association of Computing Machinery (ACM)

Member of the Association of Computing Machinery (ACM-Women)

Member of the Society for Industrial and Applied Mathematics (SIAM) and SIAM Activity Group on Supercomputing

Member of the Institute of Electrical and Electronics Engineers (IEEE)